

Engineering Mathematics: Partial Differentiation (Chain Rule)

Problem Statement

Given that $z = f(x, y)$, where x and y are functions of independent variables u and v :

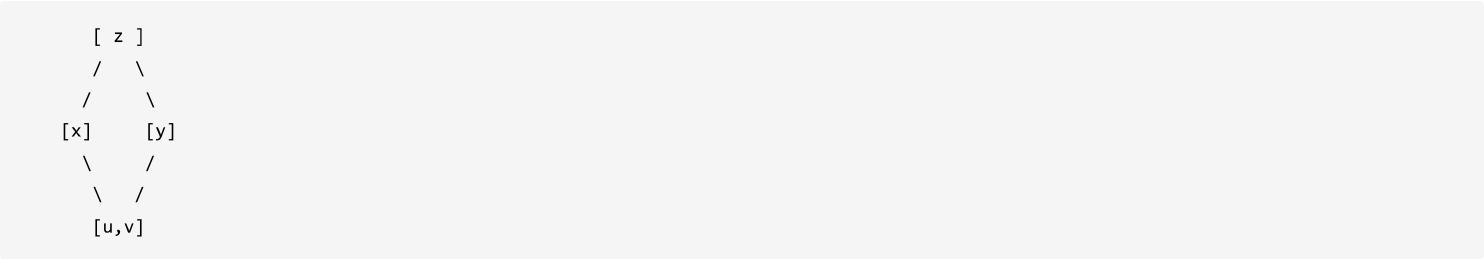
$$x = e^u \cos v$$

$$y = e^u \sin v$$

Prove that:

1. $x \frac{\partial z}{\partial v} + y \frac{\partial z}{\partial u} = e^{2u} \frac{\partial z}{\partial y}$

2. $\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = e^{2u} \left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \right]$



Prerequisites: Partial Derivatives of x and y

Before proving the statements, calculate the partial derivatives of x and y with respect to u and v .

Differentiating $x = e^u \cos v$

$$\frac{\partial x}{\partial u} = e^u \cos v = x$$

$$\frac{\partial x}{\partial v} = -e^u \sin v = -y$$

Differentiating $y = e^u \sin v$

$$\frac{\partial y}{\partial u} = e^u \sin v = y$$

$$\frac{\partial y}{\partial v} = e^u \cos v = x$$

Establishing the Chain Rule Equations

Since z depends on x and y , and both x and y depend on u and v , the total chain rule structure is:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

Substitute the known partial derivatives into these chain rule equations:

$$\frac{\partial z}{\partial u} = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \quad \text{--- (Equation 1)}$$

$$\frac{\partial z}{\partial v} = -y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} \quad \text{--- (Equation 2)}$$

Part 1: Verification of Statement (i)

We need to evaluate the Left-Hand Side (LHS):

$$\text{LHS} = x \frac{\partial z}{\partial v} + y \frac{\partial z}{\partial u}$$

Substitute Equation 1 and Equation 2 into the expression:

$$\text{LHS} = x \left(-y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} \right) + y \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right)$$

Expand the terms:

$$\text{LHS} = -xy \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} + xy \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y}$$

Cancel the counteracting terms:

$$\text{LHS} = (x^2 + y^2) \frac{\partial z}{\partial y}$$

Simplify $(x^2 + y^2)$ using the original expressions for x and y :

$$x^2 + y^2 = (e^u \cos v)^2 + (e^u \sin v)^2$$

$$x^2 + y^2 = e^{2u} \cos^2 v + e^{2u} \sin^2 v$$

$$x^2 + y^2 = e^{2u} (\cos^2 v + \sin^2 v)$$

$$x^2 + y^2 = e^{2u}$$

Substitute this value back into the expression:

$$\text{LHS} = e^{2u} \frac{\partial z}{\partial y} = \text{RHS}$$

Hence, proved.

Part 2: Verification of Statement (ii)

We need to evaluate the Left-Hand Side (LHS):

$$\text{LHS} = \left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2$$

Substitute Equation 1 and Equation 2 into the expression:

$$\text{LHS} = \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right)^2 + \left(-y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} \right)^2$$

Expand both squared brackets using $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$:

$$\left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}\right)^2 = x^2 \left(\frac{\partial z}{\partial x}\right)^2 + 2xy \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + y^2 \left(\frac{\partial z}{\partial y}\right)^2$$

$$\left(-y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y}\right)^2 = y^2 \left(\frac{\partial z}{\partial x}\right)^2 - 2xy \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + x^2 \left(\frac{\partial z}{\partial y}\right)^2$$

Add the two expanded expressions together:

$$\text{LHS} = x^2 \left(\frac{\partial z}{\partial x}\right)^2 + y^2 \left(\frac{\partial z}{\partial y}\right)^2 + y^2 \left(\frac{\partial z}{\partial x}\right)^2 + x^2 \left(\frac{\partial z}{\partial y}\right)^2$$

Group the common derivative terms:

$$\text{LHS} = (x^2 + y^2) \left(\frac{\partial z}{\partial x}\right)^2 + (x^2 + y^2) \left(\frac{\partial z}{\partial y}\right)^2$$

Factor out $(x^2 + y^2)$:

$$\text{LHS} = (x^2 + y^2) \left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \right]$$

Substitute $x^2 + y^2 = e^{2u}$:

$$\text{LHS} = e^{2u} \left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \right] = \text{RHS}$$

Hence, proved.